

FCC3 232 STRENGTH OF MATERIALS IB

CHAPTER 0 INTRODUCTION

0.1 STRENGTH OF MATERIALS

When a structural material is subjected to **applied loads**, the structural material will respond by developing **internal forces** in order to resist the applied loads. The internal forces induced in the material are expressed in terms the **stress**. **Stress** is defined as the **force per unit area of the member cross section**, and is expressed in **N/mm²**. The **strength of a material** is defined as **the stress it can suffer before failure occurs**.

“**Strength of Materials**” is a subject which develops the **techniques** used to predict the **stress induced in a structural material** when the material is subjected to applied loads, and **the manner in which the material will fail**.

0.2 TYPES OF STRESS

In structural engineering there are four types of stress which can be induced in a material: -

1. **Normal stress** is the stress due to tension or compression.
2. **Flexural stress** is the stress due to bending.
3. **Shear stress** is the stress due to shearing.
4. **Torsional stress** is the stress due to twisting.

0.3 SCOPE

FCC3 232 covers flexural stress in beams, shear stress in beams, combined stress in columns and foundations, buckling of struts, two dimensional stresses, strain energy and failure theorems.

0.4 COURSE OUTLINE

1. Flexural Stress in Beams
2. Shear Stress in Beams
3. Combined Stress
4. Buckling of Struts
5. Two-Dimensional Stress
6. Strain Energy
7. Failure Theorems

0.5 MODE OF DELIVERY

1. Soft copy of learning materials
2. Lectures: Mon 11-1 and **Fri 11-1**

0.6 MODE OF ASSESSMENT

1. 80 min **CAT** (week 9) = **30%**
2. 2 hrs **ESE** (weeks 14 and 15) = **70%**

SOME IMPORTANT FORMULAE

SECTION PROPERTIES

Rectangle $A = bd$ $I_x = bd^3/12$ $I_y = db^3/12$

Triangle $A = \frac{1}{2}bh$ $I_x = bh^3/36$ $I_y = hb^3/36$

Circle $A = \pi R^2 = \pi D^2/4$ $I_x = I_y = \pi R^4/4 = \pi D^4/64$ $J = I_x + I_y = \pi D^4/32$

Ellipse $A = \pi ab/4$ $I_x = \pi ba^3/64$ $I_y = \pi ab^3/64$ where $b > a$ and $b = x\text{-axis diam.}$

MATERIAL PROPERTIES

Mild Steel $E_s = 200 \text{ kN/mm}^2$ $G_s = 80 \text{ kN/mm}^2$ $\sigma_y = 250 \text{ N/mm}^2$

Timber $E_t = 10 \text{ kN/mm}^2$ $G_T = 5 \text{ kN/mm}^2$ $\sigma_y = 9 \text{ N/mm}^2$

Concrete $E_c = 25 \text{ kN/mm}^2$ $G_C = 10 \text{ kN/mm}^2$ $\sigma_y = 20 \text{ N/mm}^2$ (compression)

Aluminium $E_A = 70 \text{ kN/mm}^2$ $G_A = 30 \text{ kN/mm}^2$ $\sigma_y = 450 \text{ N/mm}^2$

TRIGONOMETRY

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\sin (A + B) = \sin A \cos B + \cos A \sin B$$

$$\cos (A + B) = \cos A \cos B - \sin A \sin B$$

$$\sin^2 \theta = \frac{1}{2}(1 - \cos 2\theta)$$

$$\cos^2 \theta = \frac{1}{2}(1 + \cos 2\theta)$$

$$\sin \theta \cos \theta = \frac{1}{2} \sin 2\theta$$

$$\cos^2 \theta - \sin^2 \theta = \cos 2\theta$$

$$\sin (2\theta - 180) = -\sin 2\theta$$

$$\cos (2\theta - 180) = -\cos 2\theta$$

$$d(\sin k\theta)/d\theta = k \cos k\theta$$

$$d(\cos k\theta)/d\theta = -k \sin k\theta$$

Welcome to FCC3 232
Elementary SOM Part 2